

AI Foundations for Managers



Tauhid Zaman

AI in the Core

- First time Yale SOM has put **AI in the core**
 - Using AI is now an important managerial skill.
- This is a **hands-on** course. You will use AI to **build tools** that improve real business functions.
- You guide the building, while AI handles the engineering details.



What this class covers

- **AI vibe coding**
- **AI agents**
- **Images and video**
- **Apps**
- **MCP servers**
- **Tokenomics**



| **AI vibe coding**

- **Describe what you want in words; AI writes the code**
- **You need to understand the structure and flow, i.e. the "vibe"**
- **Key skill - writing the prompt**



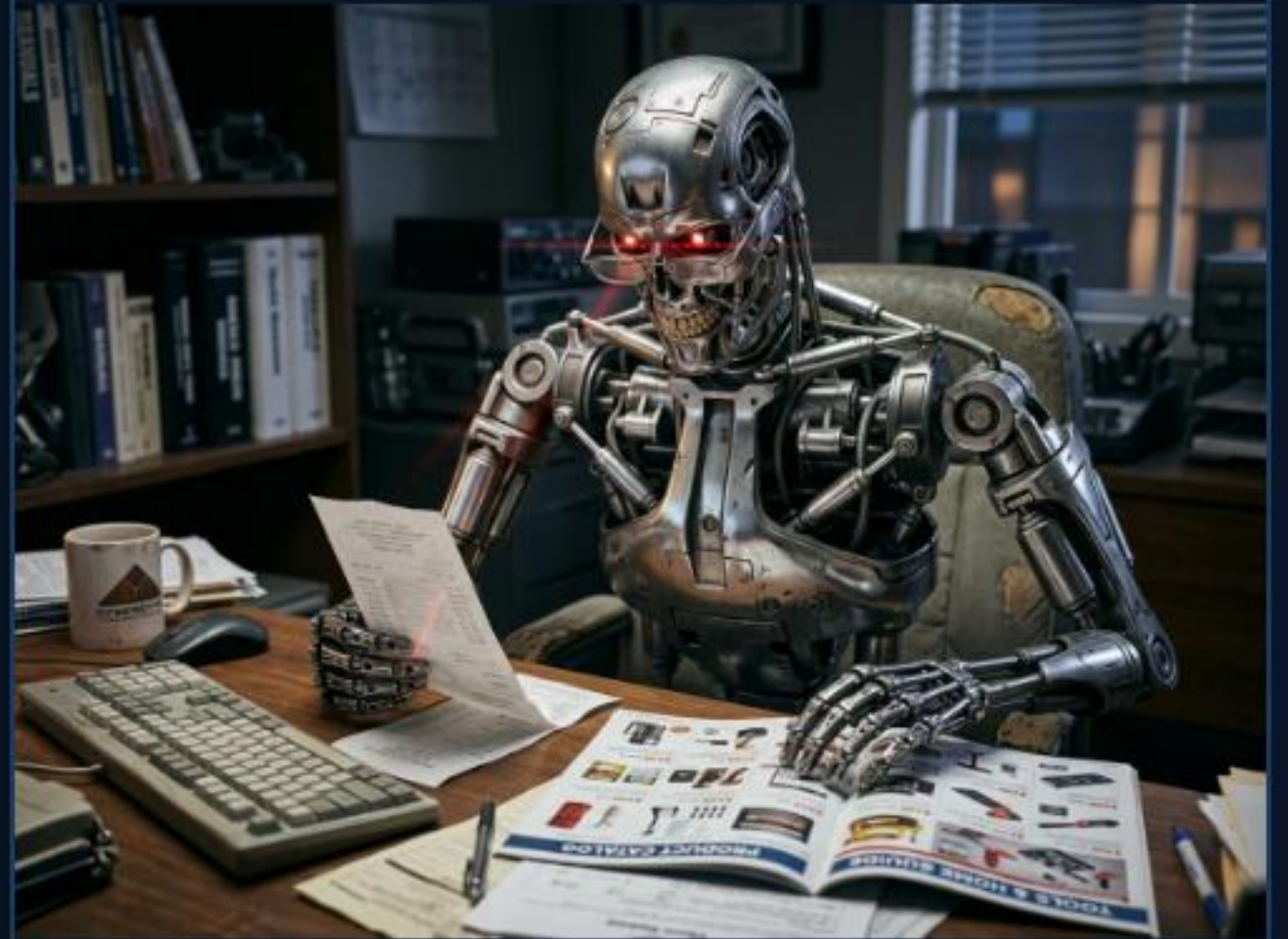
| AI agents

- **Put the model in a loop: plan → call a tool → read the result → repeat**
- **Tools: web search, read/write files, run Python, call APIs**
- **Example: find new sales prospects, design a new product line, clean up my inbox, trade stocks until I'm rich, prove the Riemann hypothesis, create a drug to cure cancer, etc.**



| Images and video

- **Vision = AI can see and *understand* images**
- **Products, people, websites, videos**
- **Build vision based tools for your agents**



Apps

- **Wrap your agent in software people can actually use**
- **Customer support chatbots, sales outreach tools, operations dashboard**
- **Web apps or mobile apps**



MCP servers

- **A standard plug-in layer so agents can call different applications via chat**
- **Gmail, Stripe, Slack, Spotify, Canvas — or build your own**
- **MCP servers are products for AI agents — not for humans.**



| Tokenomics

- **Tokens (words) → dollars: what each model call actually costs and why**
- **Design cost-effective agents — cheap model vs. smart model**
- **Guardrails and shorter prompts save money as well as risk**



Class Website

- Everything you need for the class is on the website



https://zlisto.github.io/mgt_409_fa26/

- Canvas will be used for announcements and submitting assignments

Vibe Coder

- Codex in VS Code
- Default model: gpt-5.6-luna
- Three GPT-5.6 models — pick the pig that matches the job:

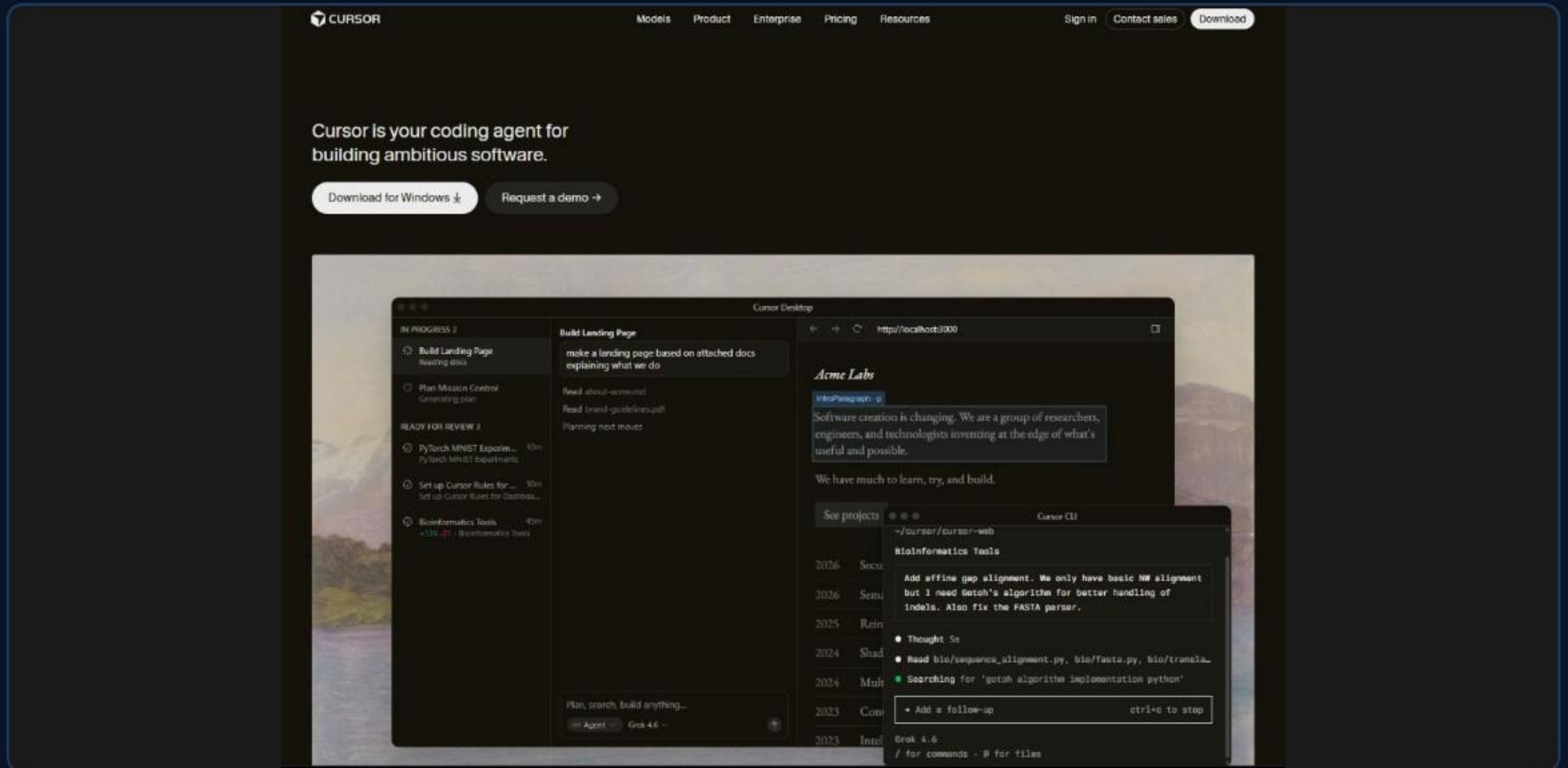


Name	Intelligence	Cost per 1M tokens (in/out)
Luna	Weakest	\$0.20 / \$1.20
Terra	Medium	\$2 / \$12
Sol	Smartest	\$5 / \$30
Yale provides you with \$40 of tokens		



Cursor

- Cursor is another vibe coder you can use (at your own expense)



Cursor Founder

- Michael Truell — CEO of Cursor



Follow

Michael Truell  

@mntruell

@cursor_ai CEO

 mntruell.com  Joined August 2014

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Before He Founded Cursor

Undergrad Interested in Your Research



Inbox x



Summarize this email



Michael Truell <truellm@mit.edu>

Wed, Oct 17, 2018, 5:03 AM



to zlisto ▾

Hello Dr. Zaman,

I am an undergrad studying 6-2 and 18. I'd be interested in talking with you about your research in Daily Fantasy Sports and potential UROP opportunities.

Would you be free to meet sometime this Friday or next week? My afternoons (after 3pm) are often free.

Best,

Michael Truell

Cursor Growth

SpaceX to acquire the AI coding startup Cursor for \$60 billion

PUBLISHED TUE, JUN 16 2026 9:06 AM EDT | UPDATED TUE, JUN 16 2026 10:51 AM EDT



Transformers



Transformers

Attention Is All You Need

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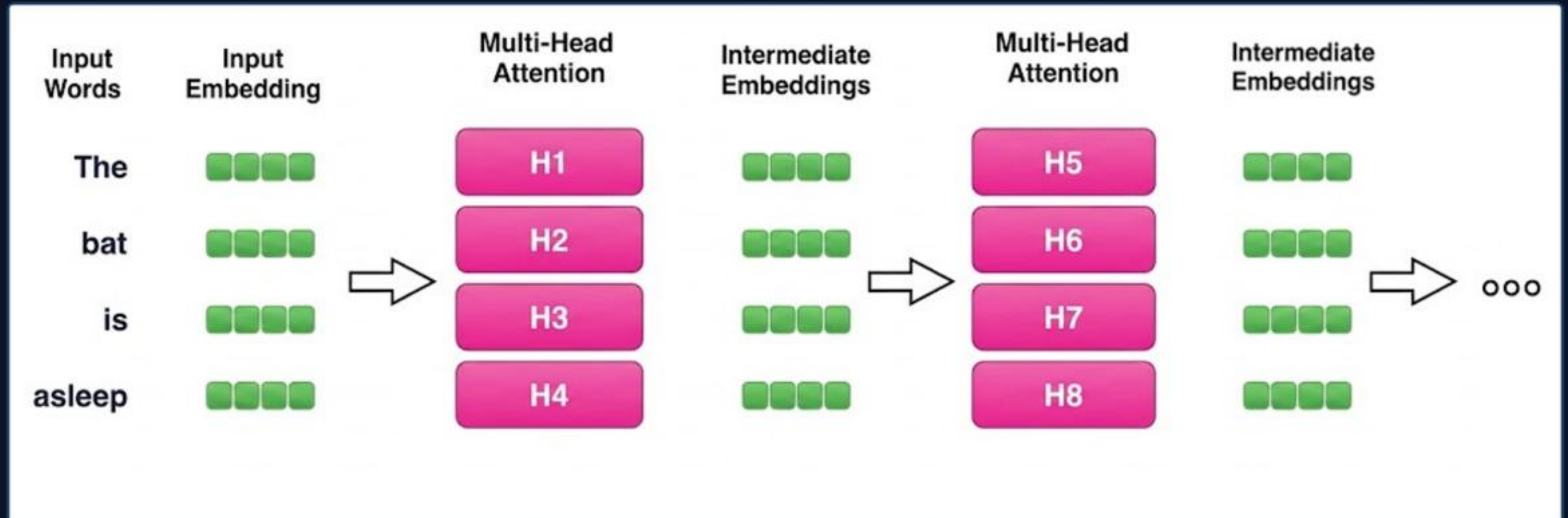
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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention

Transformer Architecture

- Each word (token) is represented as an **embedding vector**
- The transformer has **many multi-headed attention layers**
- Each layer changes the



Attention Weights

- An **attention head** computes a weight between **each pair of words**
- The weights tell you how much a word pays **"attention"** to another word

The



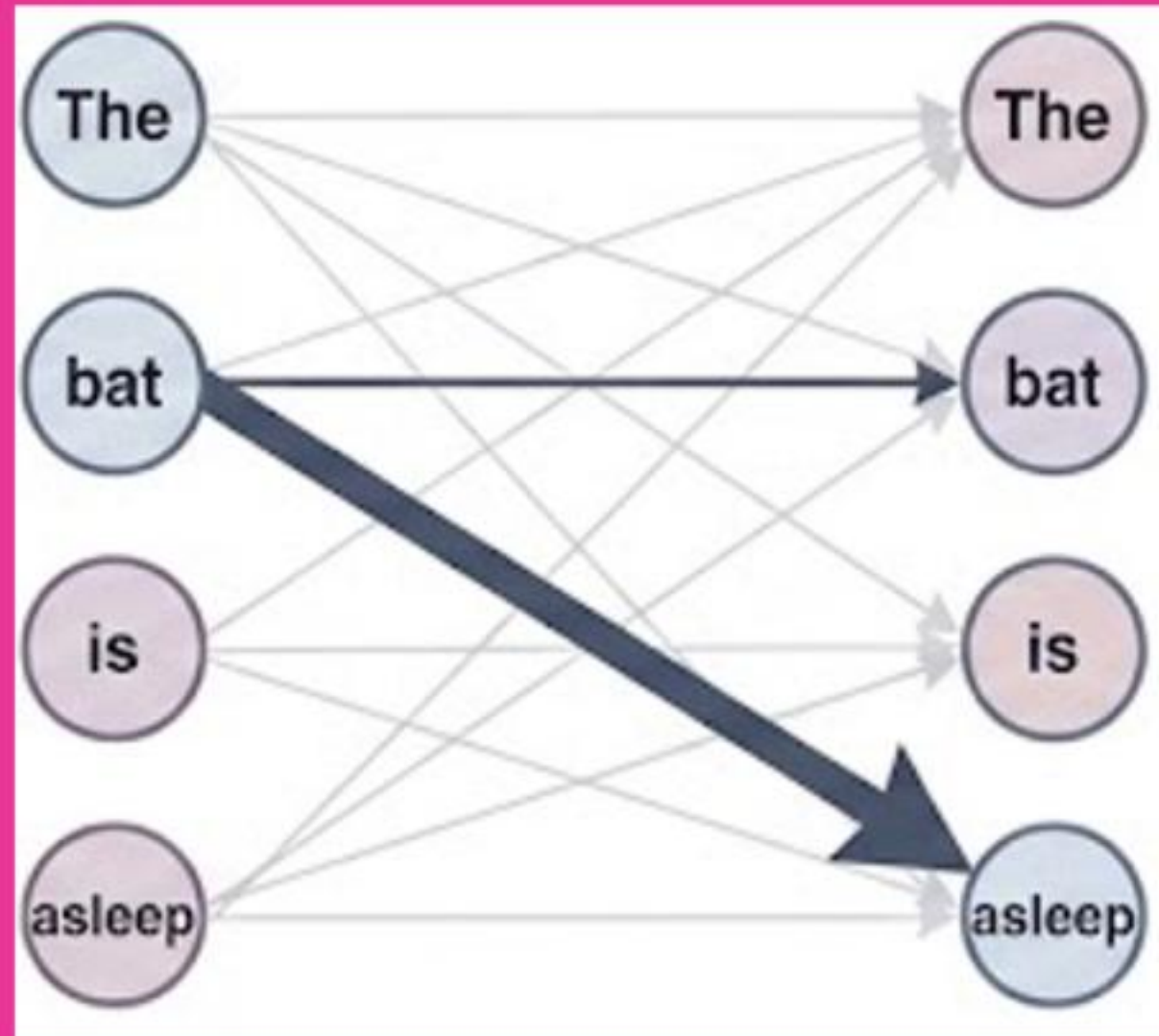
bat



is



asleep



| Context Dependent Embeddings

Consider these sentences that use the word *bat*

A **bat** flew out of the cave

The **bat** pooped on the ground

A **bat** bit me in the neck

I'm afraid of a **bat** because it is like a rat with wings

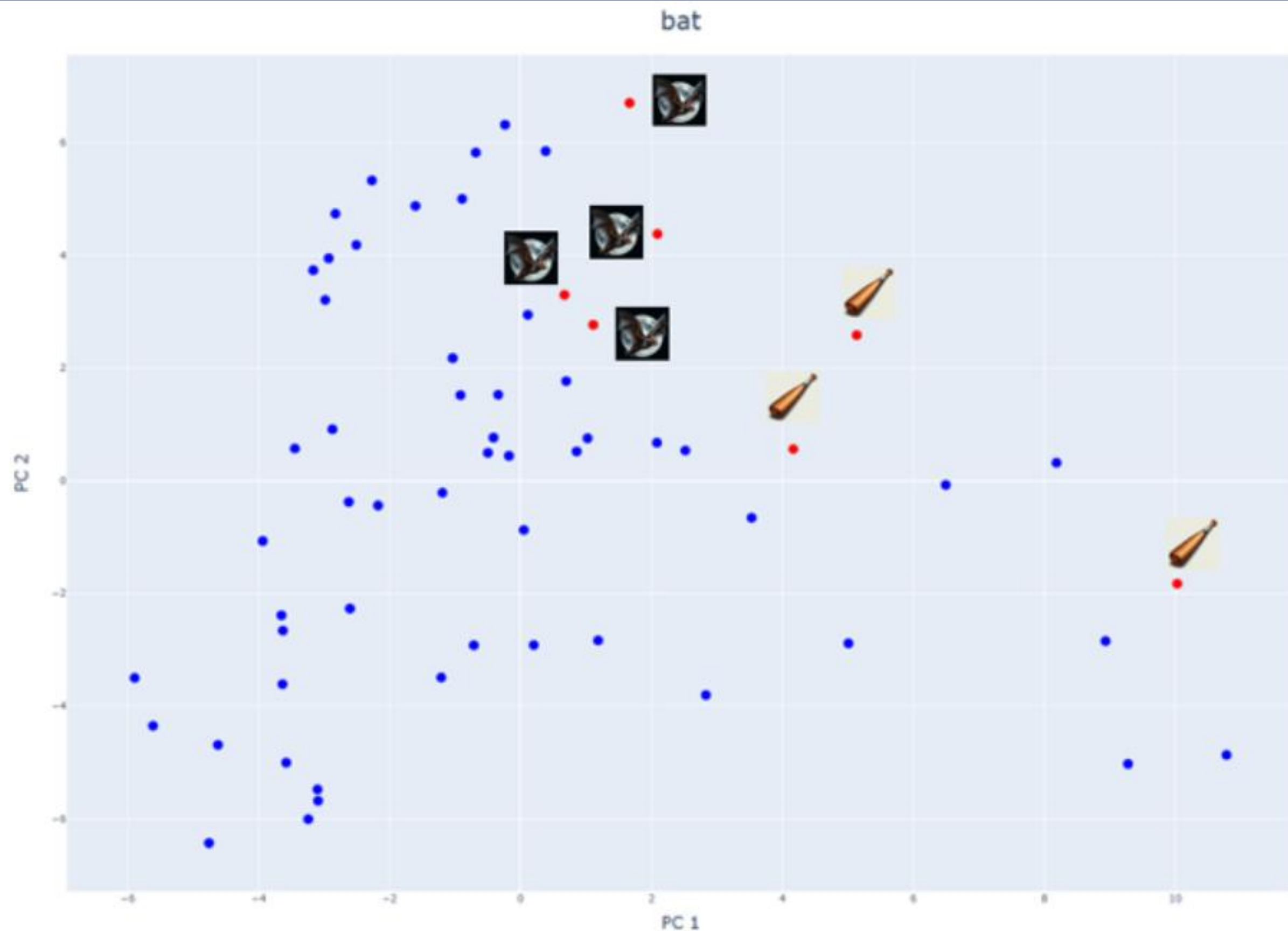
A **bat** flew out of the baseball player's hand

I hit a home run with the metal **bat**

No one swung a **bat** harder than Babe Ruth.

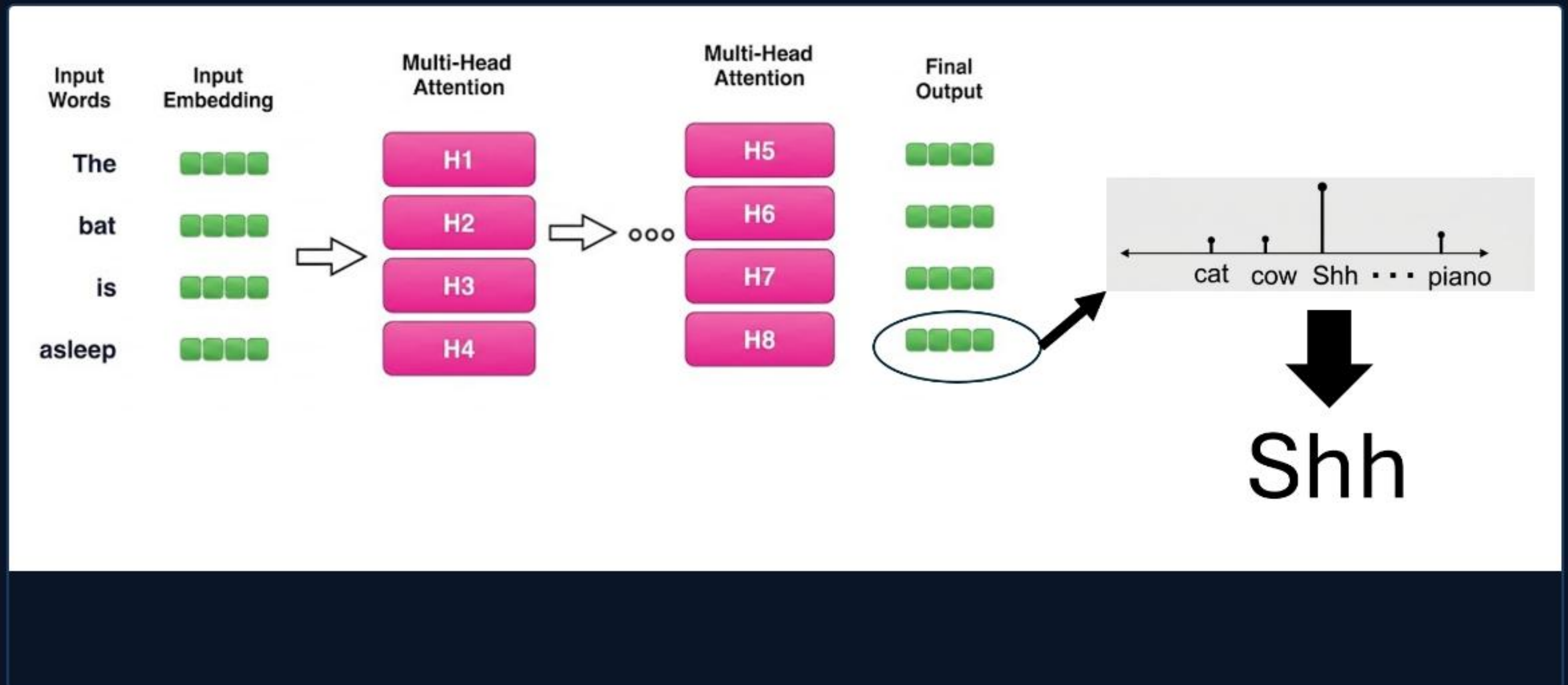


Context Dependent Embeddings



Transformers Talking

- A generative transformer turns the word embedding vector into a **probability distribution** over all words in the vocabulary
- Text is generated by **sampling** from this distribution
- These types of transformers are also called **large language models (LLMs)**



| What Can Transformers Do?

- **Generate text**
- **Measure sentiment**
- **Translate languages**
- **Summarize the reading you didn't do**
- **Answer questions**
- **Write (and debug) Python**
- **Be your friend :)**

ANYTHING!!!!

How Fast Did AI Get Smarter?

Artificial Analysis Intelligence Index vs release date — hover a point for the model name. Frontier lines for OpenAI, Anthropic, Google (Gemini), xAI (Grok), and DeepSeek (open weights).



Source: [Artificial Analysis Intelligence Index](#) (composite of hard evals: coding, science, agents, etc.). GPT-1 / GPT-2 / GPT-3 near Index ≈ 1 are illustrative (predate the Index). Other points from AA model pages & changelog (v4.1.1 where noted); index weights shift over time — use for relative progress, not exact cross-era comparisons.

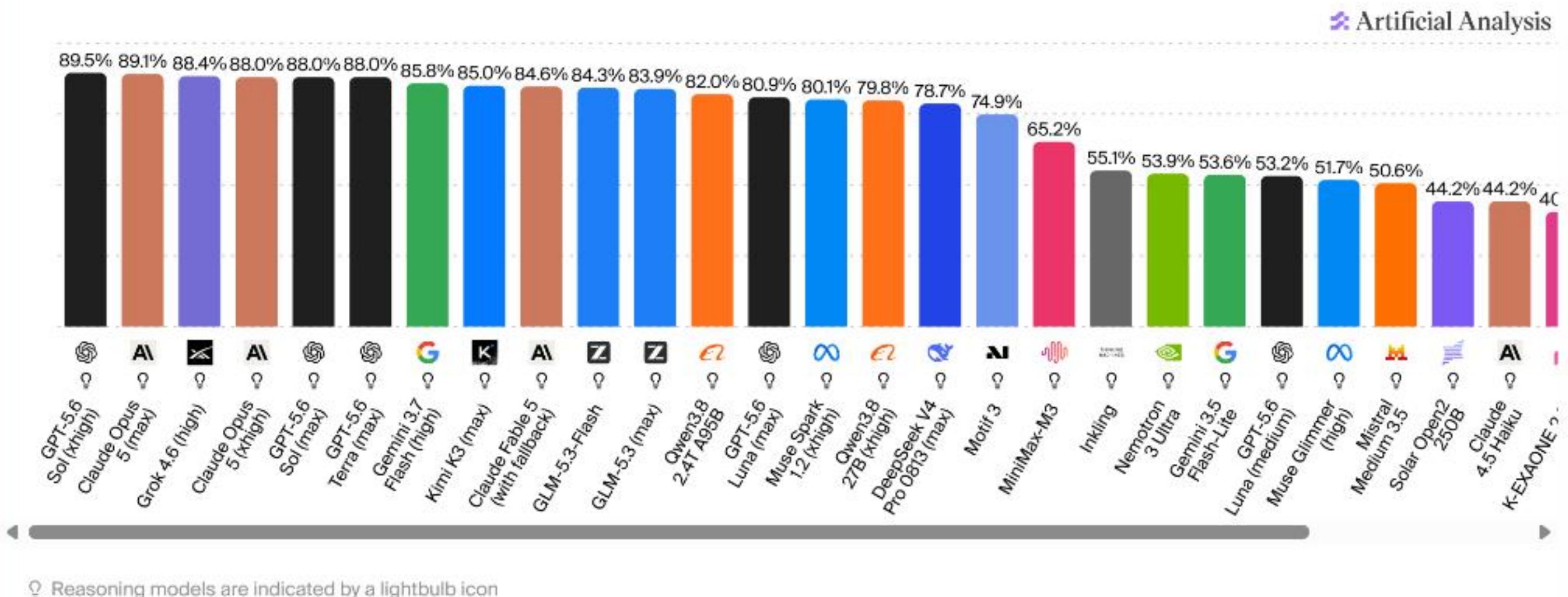
Frontier Models Intelligence

- **Terminal-Bench v2.1** tests AI agents on hard, real terminal work — 89 tasks spanning coding, sysadmin, data, training, and security
- Each task is graded by automated tests in a live shell; today's top models still miss plenty of them

artificialanalysis.ai/evaluations/terminalbench-v2-1

Terminal-Bench v2.1: Score

Benchmark developed by the Laude Institute and open-source contributors
Independently benchmarked by Artificial Analysis



Frontier Models Cost

- Same benchmark — now ranked by **average \$ per task**
- Cheaper is not always dumber; watch where the frontier sits

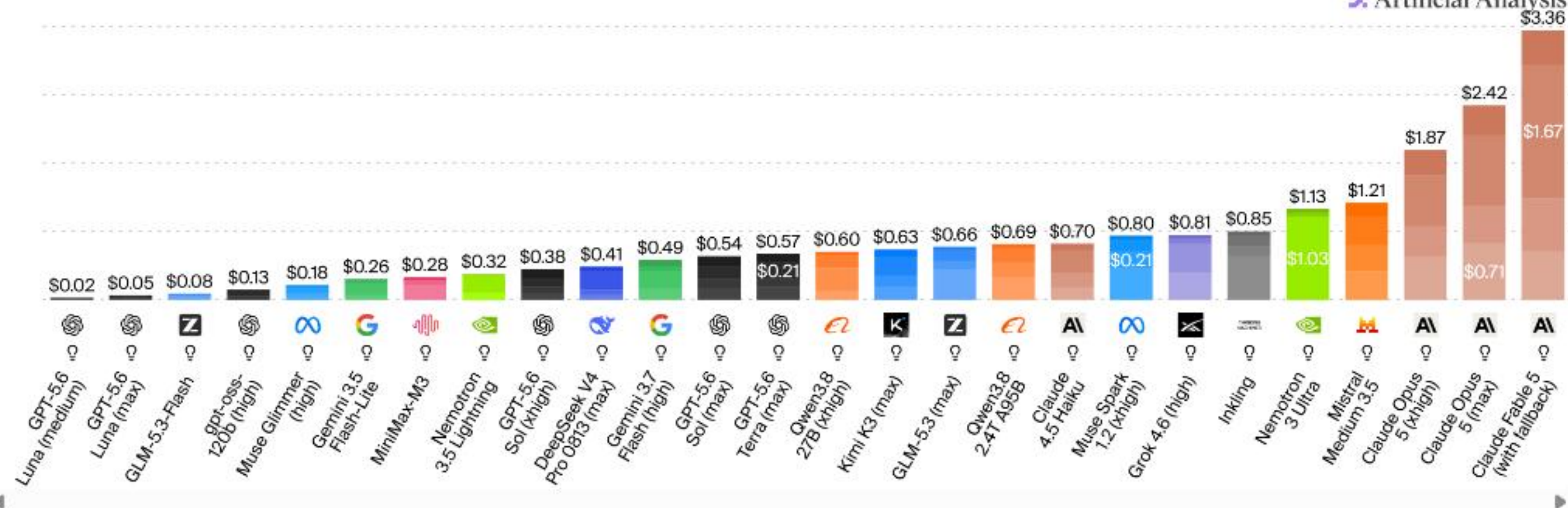
artificialanalysis.ai/evaluations/terminalbench-v2-1

Terminal-Bench v2.1: Cost per Task

Average cost per task (USD), broken down by input, cache hit, cache write, reasoning, and answer tokens

● Answer ● Reasoning ● Cache Write ● Cache Hit ● Input

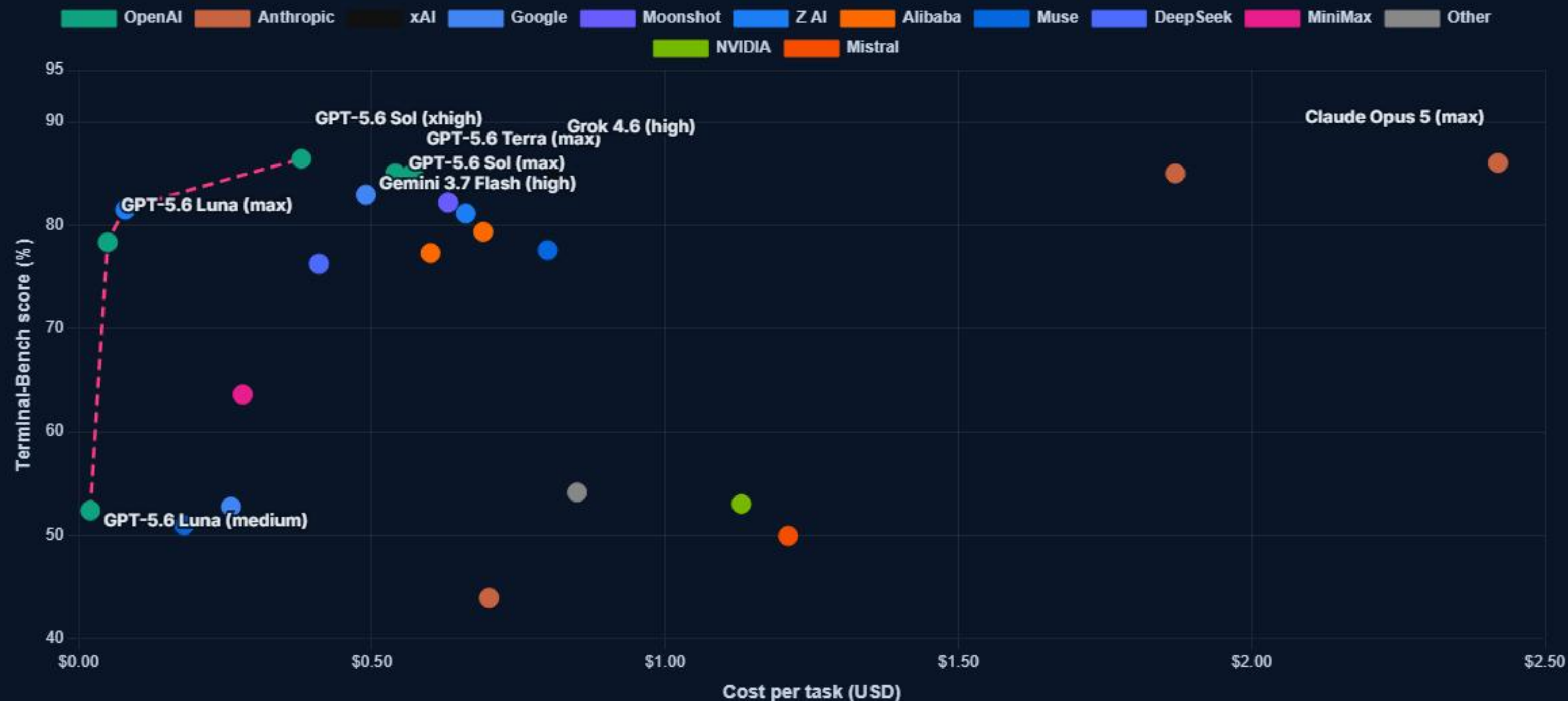
Artificial Analysis



💡 Reasoning models are indicated by a lightbulb icon

Frontier of the Frontier

Terminal-Bench v2.1 — cost per task vs score. Hover a dot for the model. The dashed line is the cheap-and-smart frontier (nothing cheaper scores higher).



Numbers read from Artificial Analysis Terminal-Bench v2.1 charts (27 Aug 2026). Source: artificialanalysis.ai/evaluations/terminalbench-v2-1

When Humans Meet Machine Intelligence



Garry Kasparov vs Deep Blue (1997)



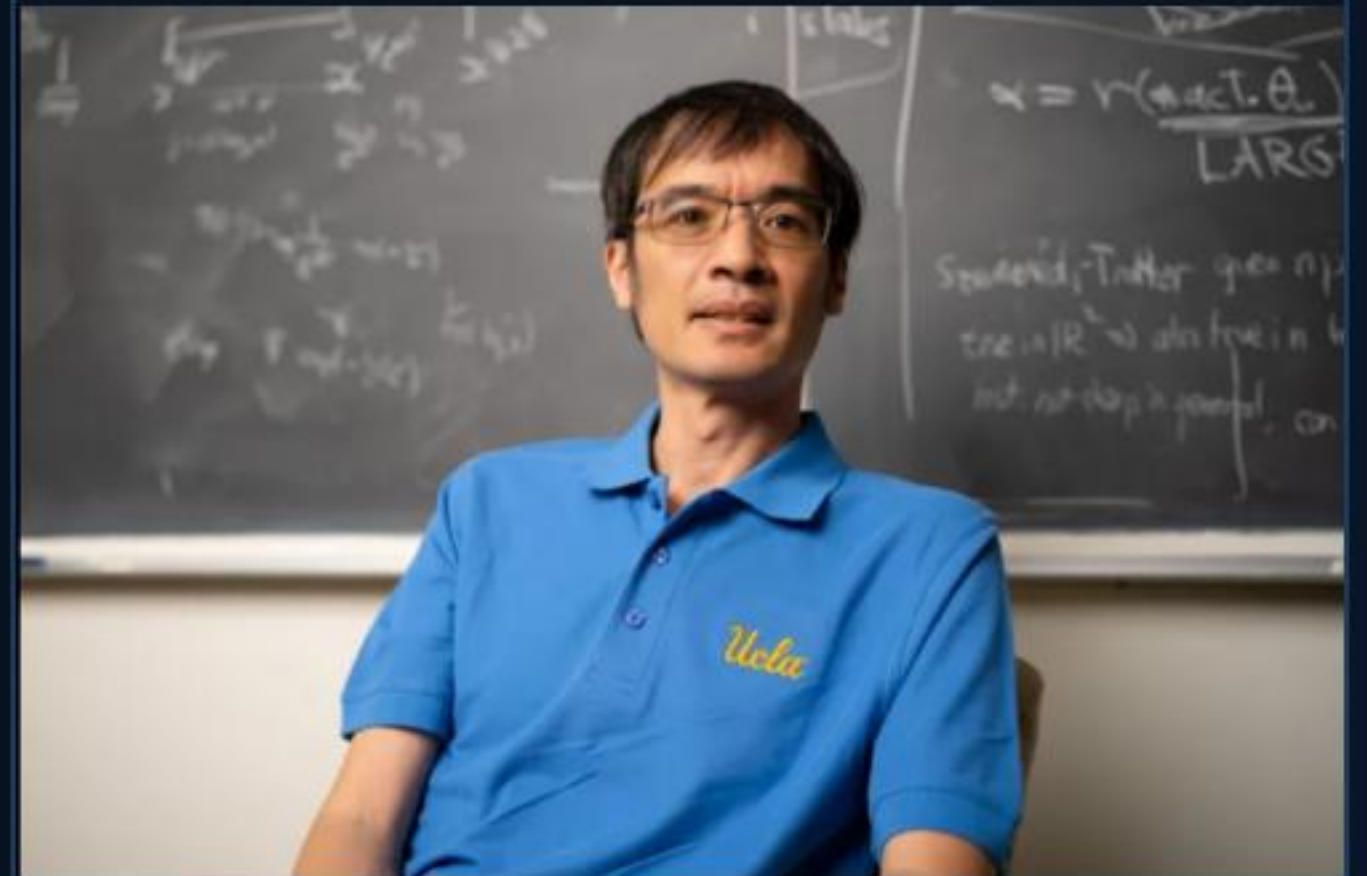
Who is Stoker?
(I FOR ONE WELCOME OUR
NEW COMPUTER OVERLORDS)

Ken Jennings vs Watson (2011)

- Kasparov lost — and was furious: accused IBM of cheating, stormed off the cultural high ground
- Jennings lost too — but accepted defeat with **grace and humor** ("I welcome our new computer overlords")
- Same shock, opposite response — the winners are the ones who **partner with the machine**

| Terence Tao on AI and Mathematics

- **Terence Tao — Fields medalist, often called the greatest living mathematician**
- **He believes AI can advance mathematics — not replace human insight**
- **Math must rethink the field and embrace AI, or lose the narrative**
- [New Scientist interview](#)



Recent AI Achievements in Math

• OpenAI (Astra) — 10 long-open problems

- Non-sofic group
- Connes rigidity conjecture
- Erdős Problem #183
- Ehrhart volume conjecture
- Sphere-packing bound + 5 more

• Anthropic (Claude)

- Riemann zeta zeros — $> \frac{2}{3}$ on the critical line
- S^6 complex structure (Hopf problem, open since 1948; still under review)



AI-Maxxing Frame

- Once you embrace AI and decide to **AI-maxx**, you're in a new frame — you stop bolting AI onto the old workflow
- Not “use AI to fill in a spreadsheet” → **no spreadsheets**; AI updates your records database directly
- Not “use AI to skim resumes in HR” → AI interviews every candidate in depth and maps real strengths
- Not “use AI to draft customer-support replies” → AI owns the resolution loop and only escalates true edge cases



| New Tech, New Paradigm

- Every transformative technology starts by bolting onto the old world — until someone sees what the tool is *really* for
- Early cameras just filmed stage plays — then someone invented **cinema**
- AI today is that kind of leap: design the next paradigm, not a faster version of yesterday



Course logistics

10%

Attendance

40%

Homework

50%

Final project

- **6 homeworks**
 - Vibe code solutions to business problems
 - Submit code in zip files onto Canvas (later as online GitHub repos)
 - Anti-cheating measures in homeworks - don't blindly give the problem to the AI. Read it, understand it, and interact with the AI to complete it
- **Final project**
 - Business AI Co-Pilot that integrates the class modules
 - Details will be provided soon on website
- **Attendance**
 - Attendance is mandatory for class
 - Attendance is taken at the beginning of class via QR code - don't be late
 - If you miss attendance, email the professor and explain your reasons
 - Tell the professor to put up the code now

| Next Class

- We will be **vibe coding**
- Setup VS Code, Codex, and Portkey
- Bring ideas for things you want to build
- We might try them in class *for fun*

